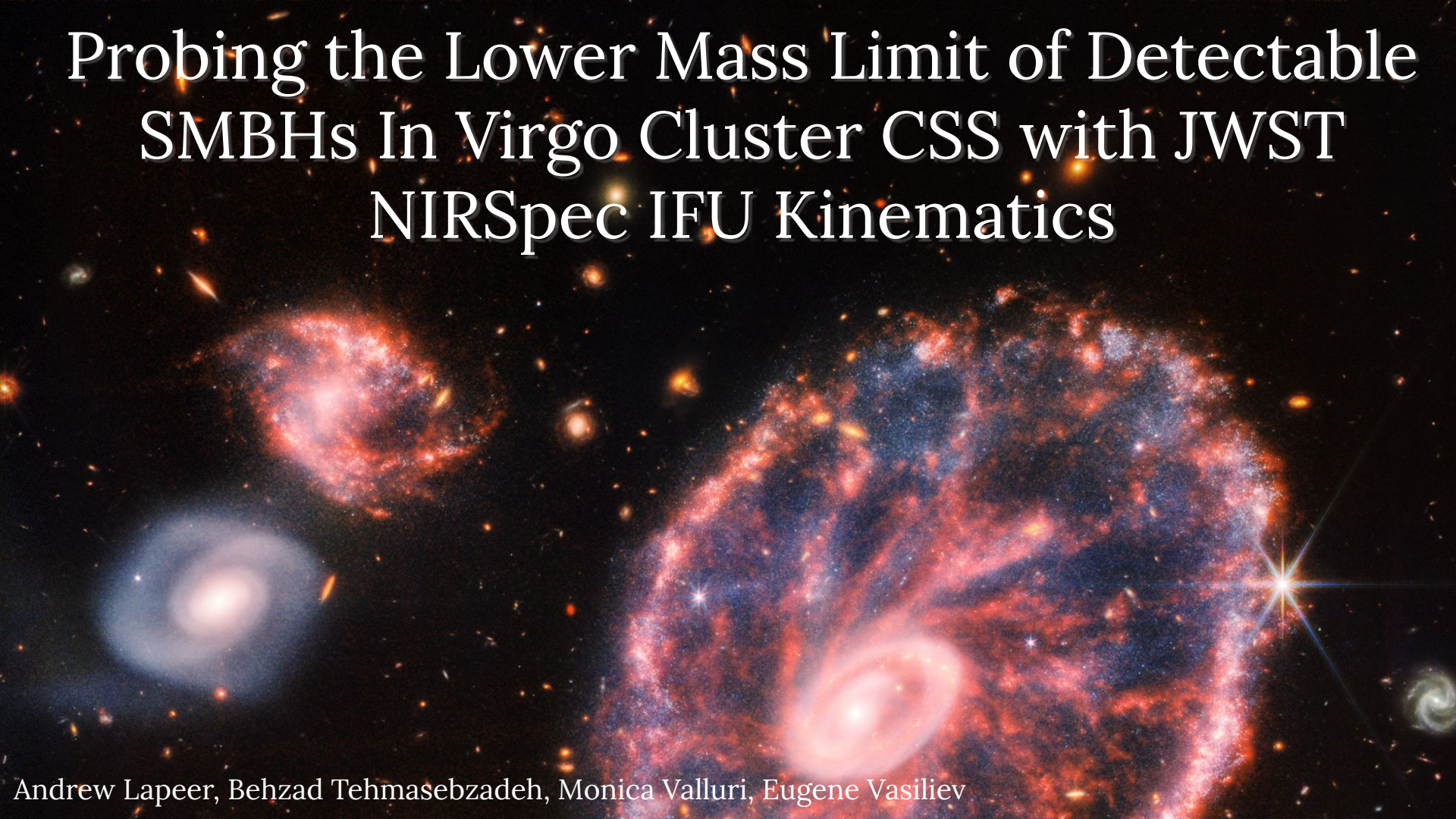
Probing the Lower Mass Limit of Detectable SMBHs In Virgo Cluster CSS with JWST NIRSpec IFU Kinematics



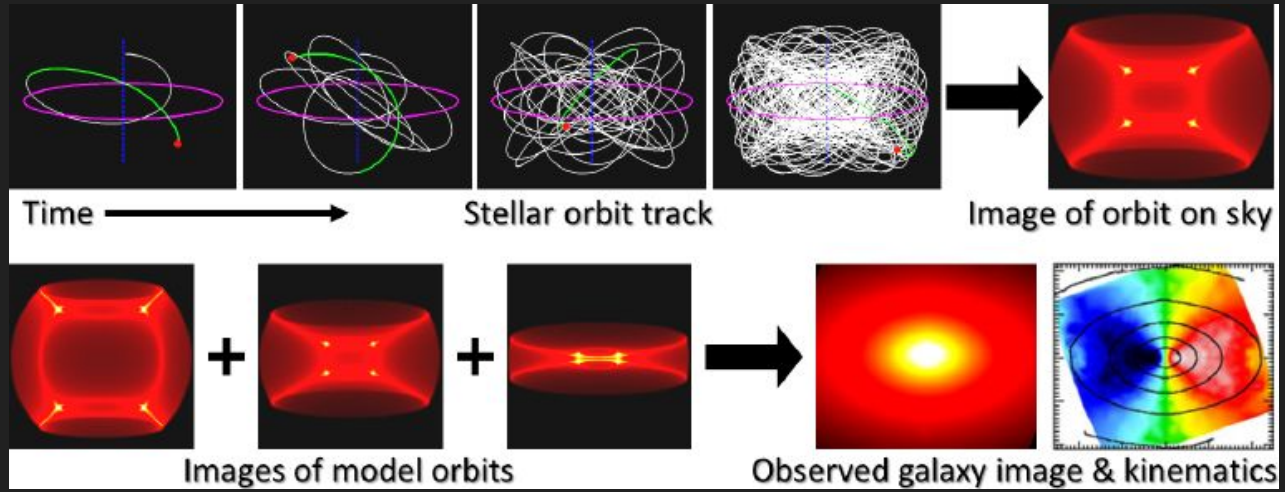
Andrew Lapeer, Behzad Tehmasebzadeh, Monica Valluri, Eugene Vasiliev

Schwarzschild Modeling - The Orbit-Superposition Method

The most prominent dynamical mass determination method

Utilizes the orbital structure of galaxies to construct models in three-integral space

Models can be used to determine mass of dark components (M_{Halo}, M_{BH}), velocity anisotropy, bar pattern speed, and more!



The General Idea for M_{BH} constraint:

Observe the motions of stars surrounding SMBH

↓

Use Gauss-Hermite parameterization of LOSVDs to obtain σ, h3 and h4 moments, and higher order moments

↓

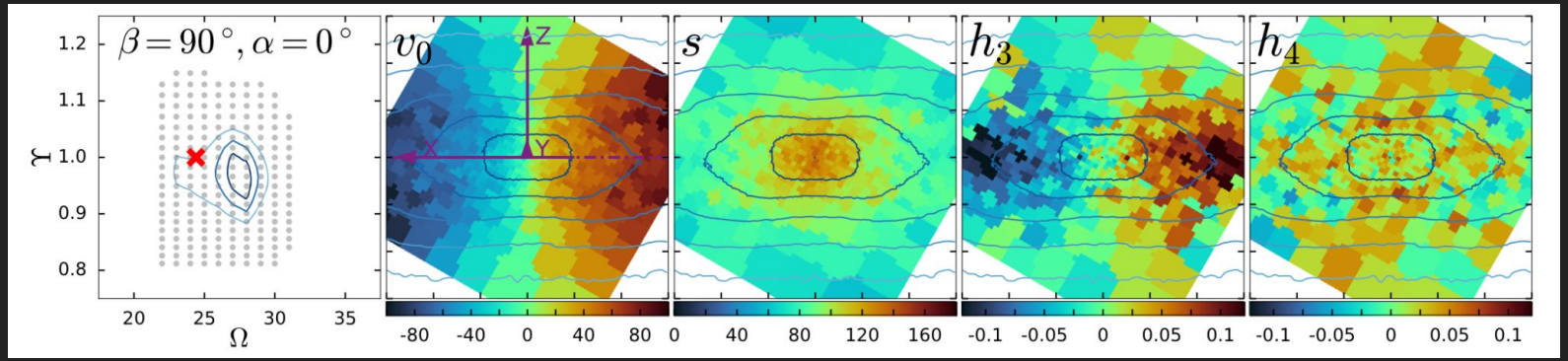
Using observational constraints, iteratively solve Poisson equation to find a potential whose DF best fits the observations

↓

Use the best fit DF to constrain the dark mass component required to account for observed kinematics

FORSTAND: Flexible orbit superposition toolbox for analyzing dynamical models

- Newest implementation of the Schwarzschild technique
  - Generalized
    - Applicable to all morphological types
  - Highly optimized (computationally efficient)
  - Vasiliev & Valluri 2020
- Built upon the AGAMA (Vasiliev 2019) galactic dynamics framework



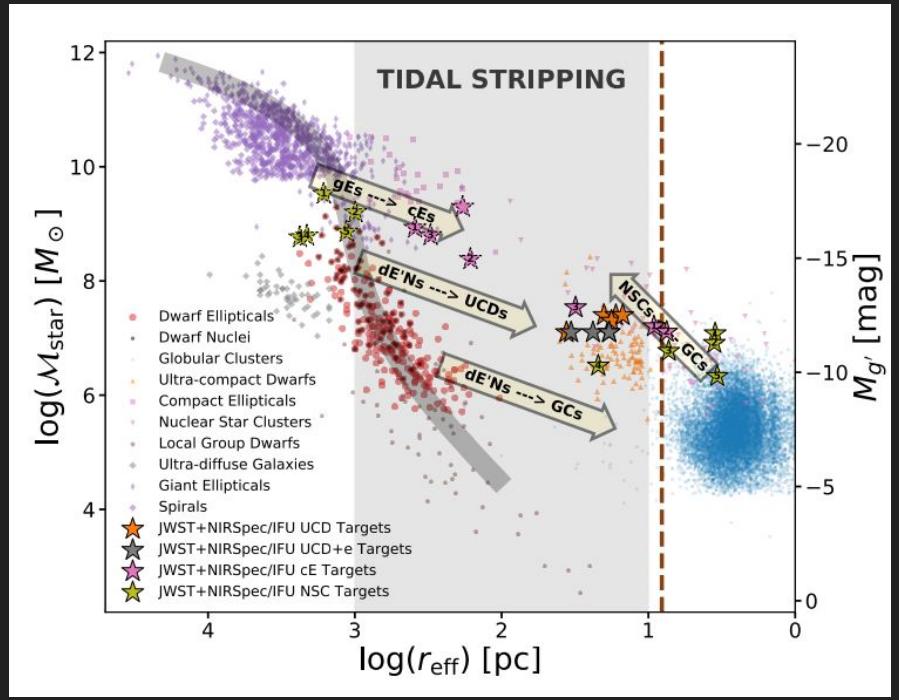
JWST Proposal 2567 - Do Massive Black Holes Come in Small Packages? A census of black holes in compact stellar systems in the Virgo cluster. PI: M. Taylor (UCalgary).

- JWST NIRspec IFU observations of 18 Virgo cluster (all observations received as of May!) compact stellar systems (i.e., really tiny stellar systems!)

- Robust testing grounds for CSS formation and SMBH seeding mechanisms!

- But what results can we reasonably expect to obtain?

Generating realistic mocks to represent JWST target sample



$$\rho = \rho_0 \left(\frac{\tilde{r}}{a}\right)^{-\gamma} \left[1 + \left(\frac{\tilde{r}}{a}\right)^{\alpha}\right]^{\frac{\gamma-\beta}{\alpha}} \times \exp\left[-\left(\frac{\tilde{r}}{r_{\rm cut}}\right)^{\xi}\right]$$

Two free parameters (γ and a)

↓

Make many mocks over range of suitable γ and a, fix everything else

↓

Project each N-body snapshot, fit 2D Sérsic to find n an r_e

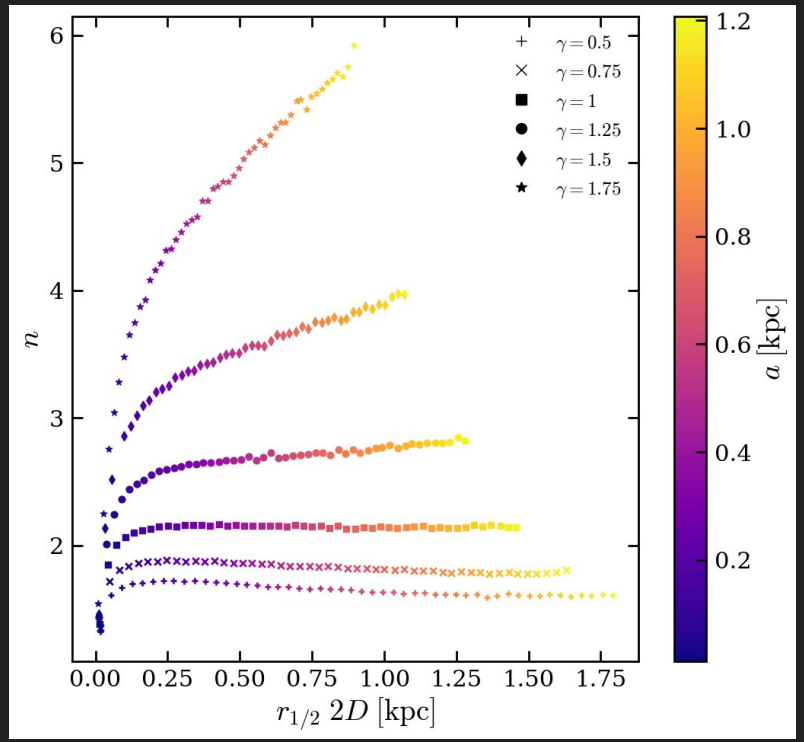
↓

Use observable quantities (n an r_e) to obtain non-observable quantities (γ and a) for each target classification

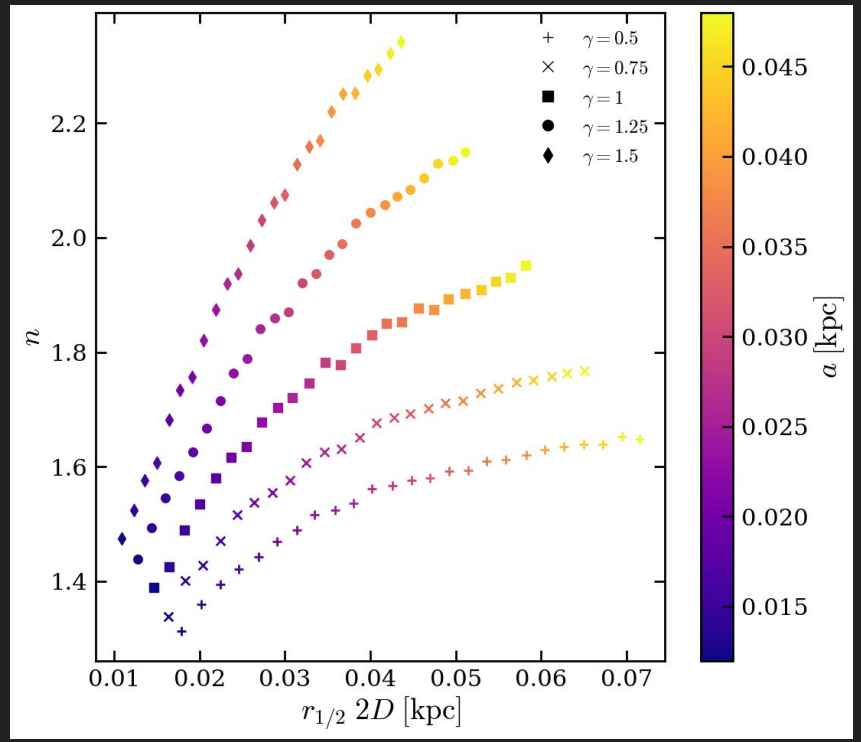
↓

Construct mocks based on determined γ and a

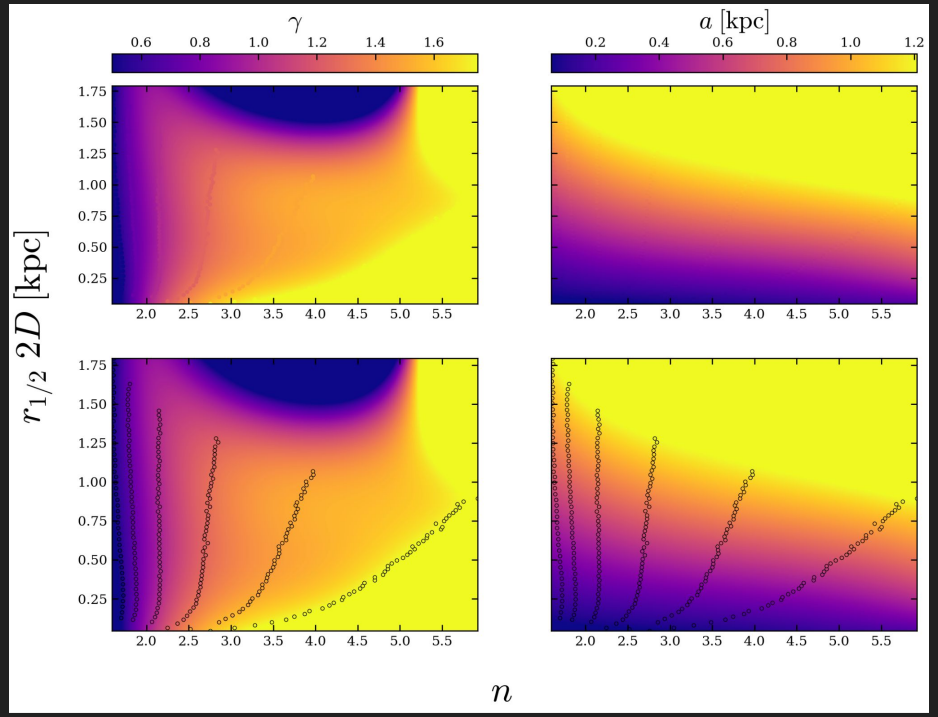
cEs


$\gamma = 0.5$
$\gamma = 0.75$
$\gamma = 1$
$\gamma = 1.25$
$\gamma = 1.5$
$\gamma = 1.75$
n
$r_{1/2}$ $2D$ [kpc]
a [kpc]

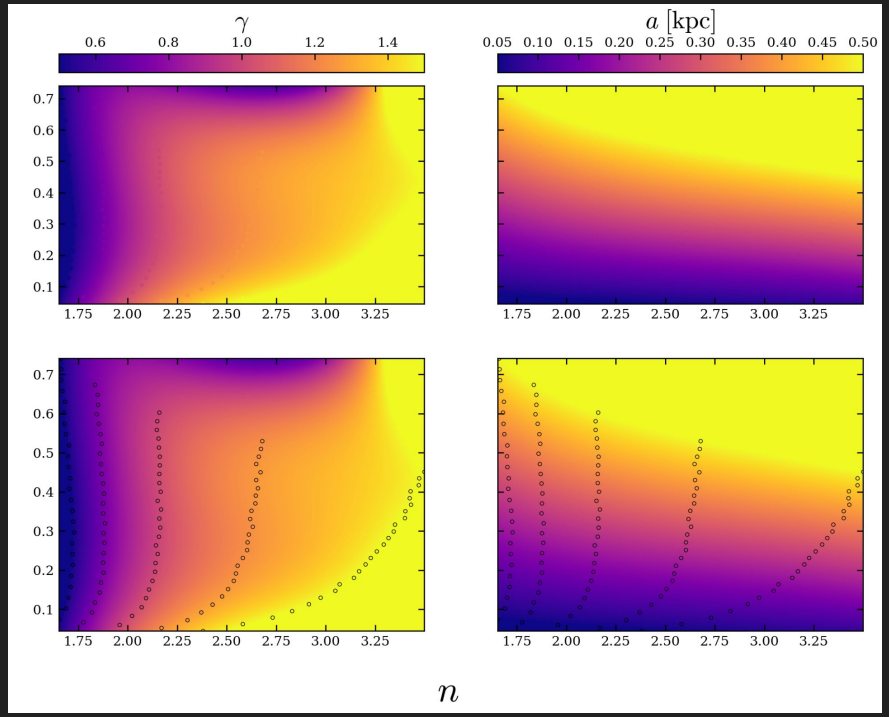
UCDs


$\gamma = 0.5$
$\gamma = 0.75$
$\gamma = 1$
$\gamma = 1.25$
$\gamma = 1.5$
n
$r_{1/2}$ $2D$ [kpc]
a [kpc]

cEs


γ
0.6 0.8 1.0 1.2 1.4 1.6
a [kpc]
0.2 0.4 0.6 0.8 1.0 1.2
r1/2 2D [kpc]
1.75 1.50 1.25 1.00 0.75 0.50 0.25
2.0 2.5 3.0 3.5 4.0 4.5 5.0 5.5
n

UCDs


γ
0.6 0.8 1.0 1.2 1.4
a [kpc]
0.05 0.10 0.15 0.20 0.25 0.30 0.35 0.40 0.45 0.50
0.7 0.6 0.5 0.4 0.3 0.2 0.1
1.75 2.00 2.25 2.50 2.75 3.00 3.25
n

With accurate mocks, we can then…

Construct accurate 3D representations of our targets

↓

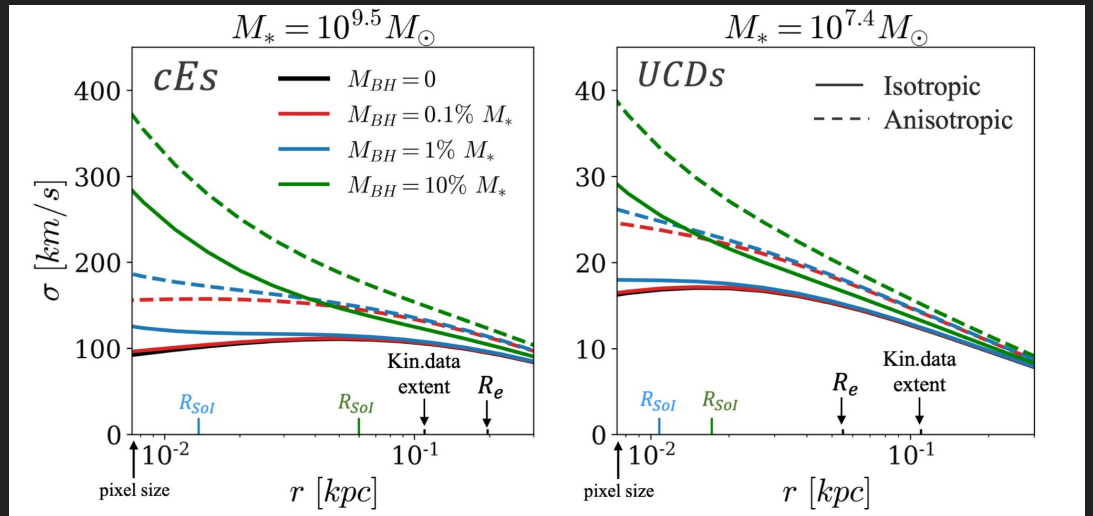
Project those mocks and "observe" them

↓

Use Schwarzschild modeling with mock kinematics, which spatially and spectrally mimic JWST

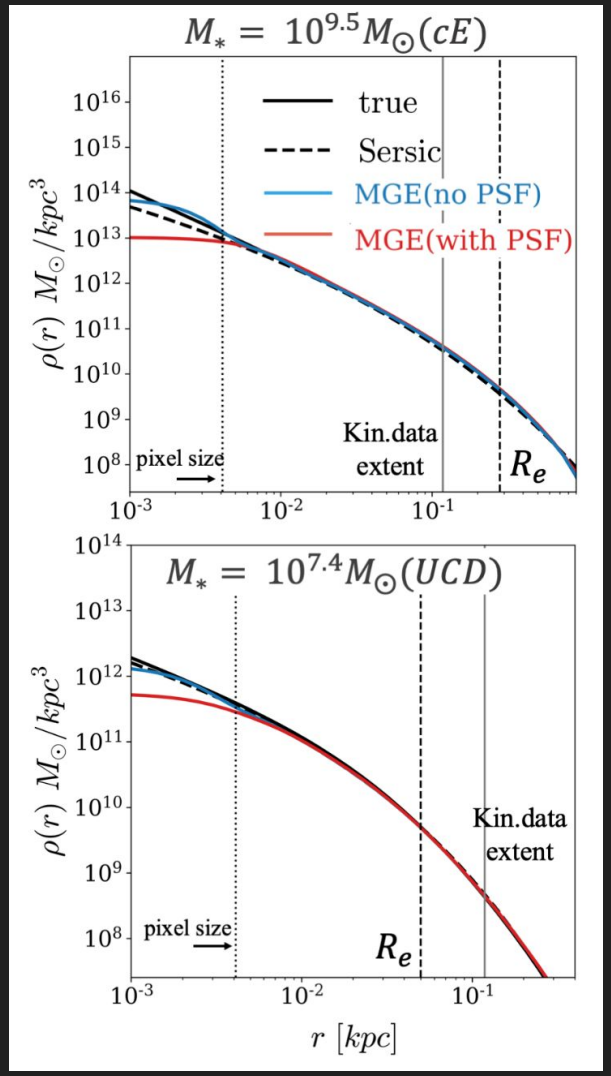
↓

Probe the lower limit of detectable SMBH mass for a range of cases!



Deprojection

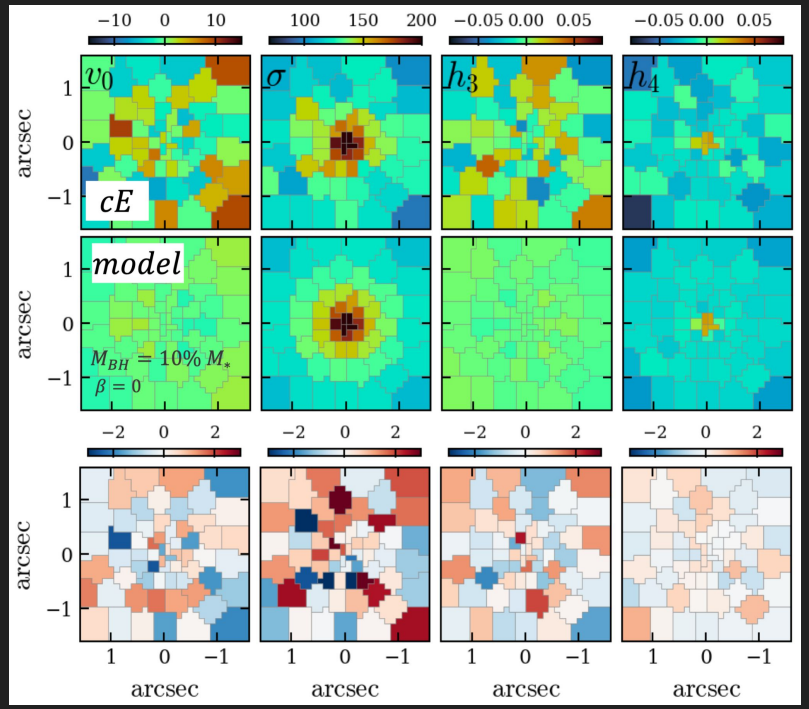
Photometry		3D density
Multi-Gaussian Expansion parameterization (MgeFit)	→	Constructing 3D MGE models
Sersic parameterization (GALFIT)	→	Numerically computing the spherical deprojection integral (AGAMA)



Compact Elliptical Type Galaxies

$M_* \cong 10^8 - 10^9\ M_{sun}$

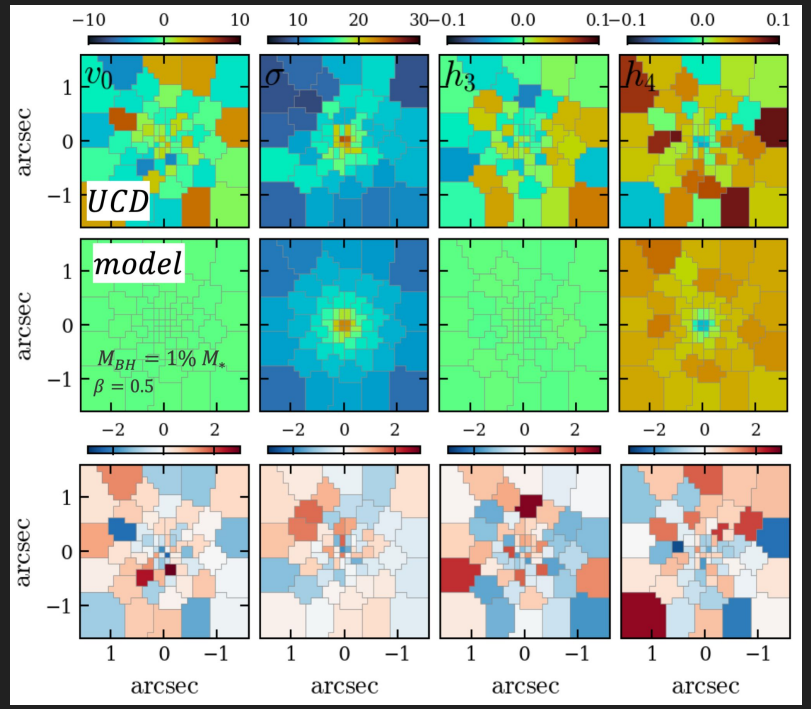
$M_{BH} = 0.10\ M_*$



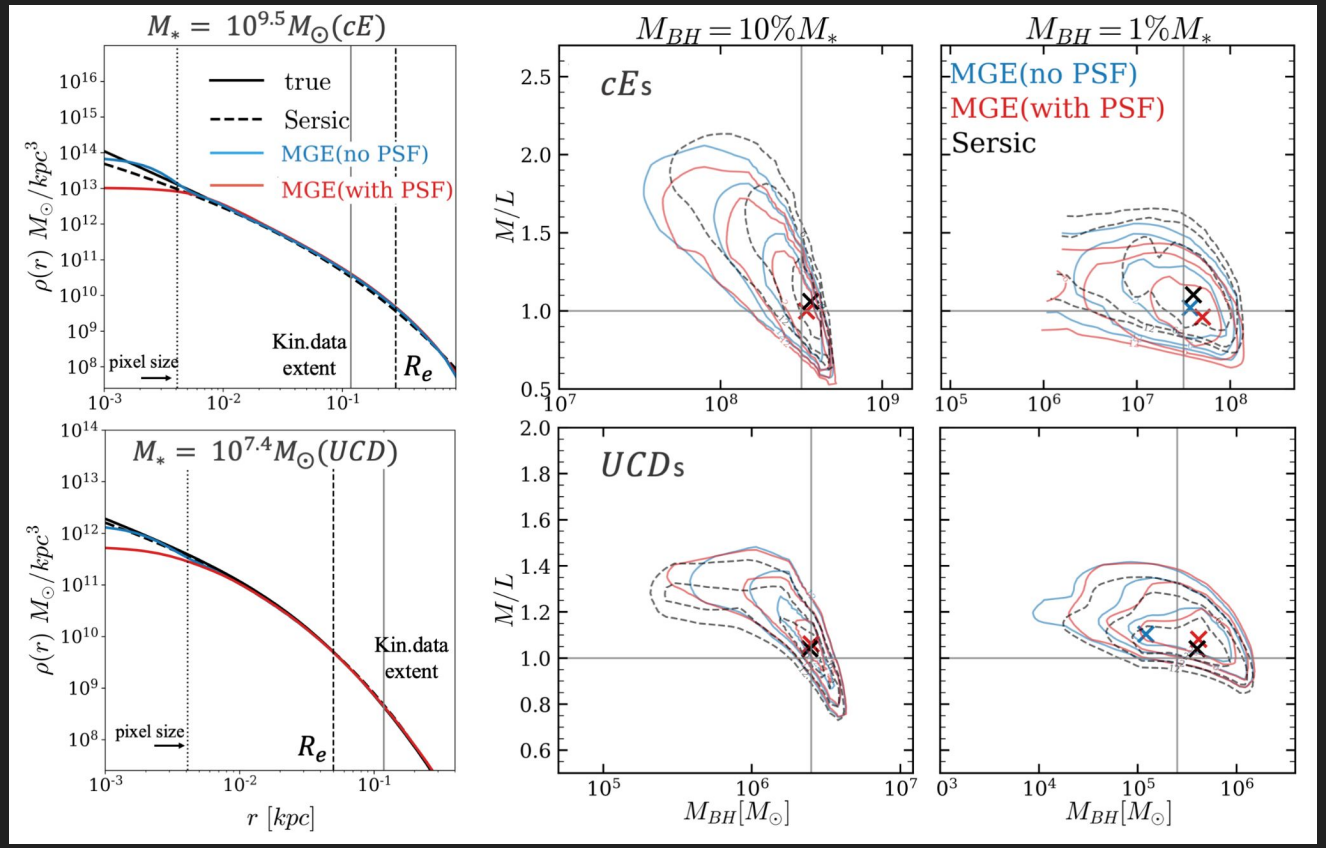
Ultracompact Dwarf Type Galaxies

$M_* \cong 10^{7.5}\ M_{sun}$

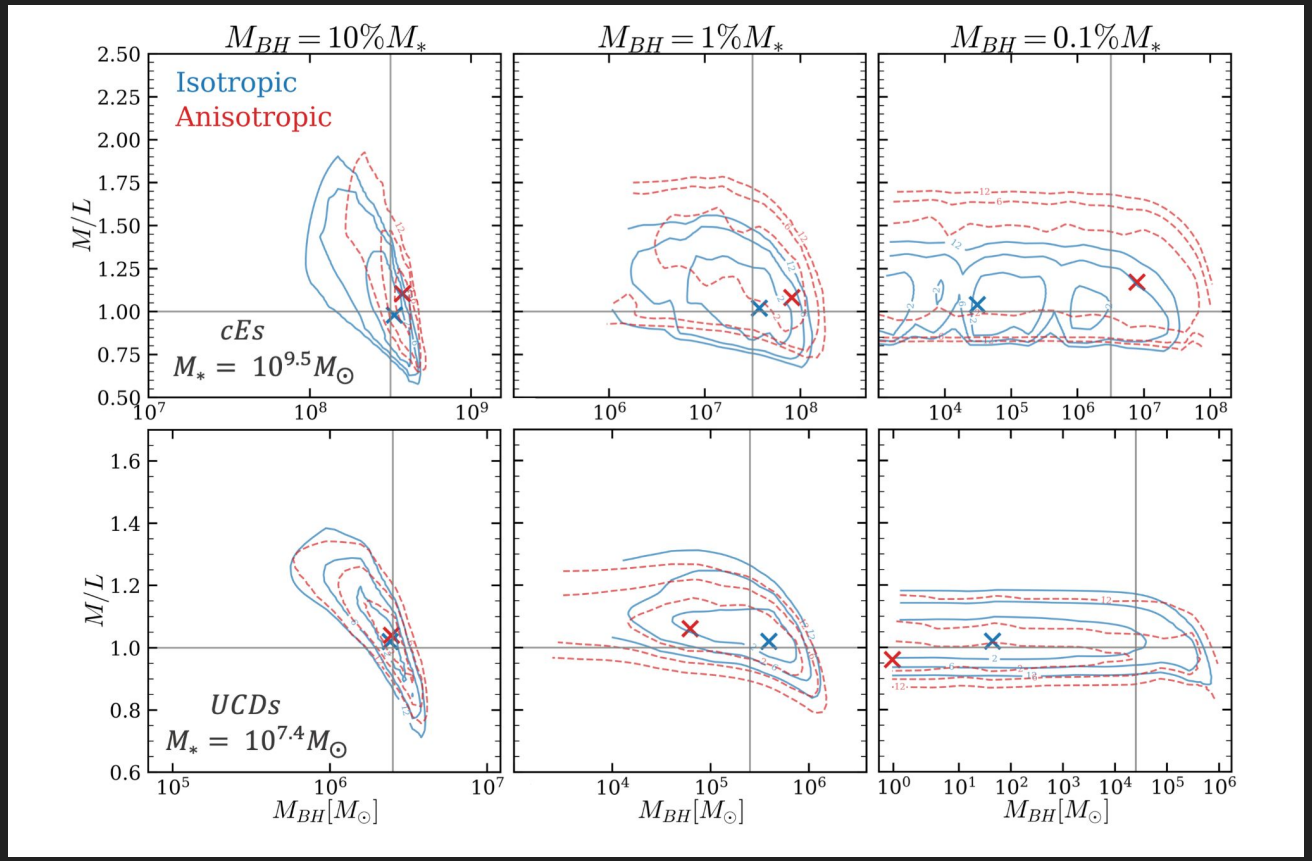
$M_{BH} = 0.01\ M_*$



Comparing Deprojection Methods



Conclusion: Detection limit of ~1%M_*



Whats next?

- Convolve projected mock images with PSF

- Generate mocks for NSC, r_e~10 pc (~ 2 spaxels)
  - Multi-component NSC+host galaxy

- Try N-body simulations (evolved) for UCD+BH and NSC+BH to ensure that our mocks are realistic (i.e. they don't evolve rapidly to different distributions)

- Different realization of noise

Interested in learning more?

Results from this project will be published late-2023/early-2024

Taylor et al. in prep, Tehmasebzadeh & Lapeer et al. in prep.

Always happy to chat more! : D

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